

Nash's Problem

eight curl-free fields, and why there are exactly eight

problem. find $X \subset \mathbb{R}^3$ such that, on $U = \mathbb{R}^3 \setminus X$, the space V of continuously differentiable vector fields with $\nabla \times F = 0$, modulo the space W of gradient fields $F = \nabla f$, has $\dim(V/W) = 8$.

construction. remove eight parallel lines:

$$L_j = \{(j, 0, z) : z \in \mathbb{R}\}, \quad X = \bigcup_{j=1}^8 L_j.$$

for each j , define on U

$$F_j(x, y, z) = \frac{1}{2\pi} \left(-\frac{y}{(x-j)^2 + y^2}, \frac{x-j}{(x-j)^2 + y^2}, 0 \right).$$

each F_j is smooth on U , independent of z , and a direct differentiation gives $\nabla \times F_j = 0$. locally, F_j is $\nabla \theta_j / (2\pi)$, where θ_j is the angle around L_j ; globally, that angle changes by 2π after one turn.

eight independent classes. let $\gamma_i(t) = (i + \varepsilon \cos t, \varepsilon \sin t, 0)$ for $0 \leq t \leq 2\pi$ and $0 < \varepsilon < 1/3$. a direct parametrization gives $F_i(\gamma_i(t)) \cdot \gamma_i'(t) = 1/(2\pi)$. if $j \neq i$, the disk bounded by γ_i misses L_j , so stokes' theorem gives

$$\oint_{\gamma_i} F_j \cdot d\mathbf{r} = \delta_{ij}.$$

every gradient has zero integral around a closed curve. hence no nonzero linear combination of the F_j is a gradient: their eight classes in V/W are independent.

why there are no others. sliding points vertically to $z = 0$ retracts U onto the plane with the eight points $(j, 0)$ removed. every closed curve in this punctured plane is, for purposes of integrating a curl-free field, a sum of loops around those eight punctures. equivalently, its first homology group is $\mathbb{Z}^8$, generated by the γ_j .

now take any $F \in V$ and set

$$a_j = \oint_{\gamma_j} F \cdot d\mathbf{r}, \quad G = F - \sum_{j=1}^8 a_j F_j.$$

the displayed circulation identity makes every period of G around γ_j zero. since these loops generate all periods, $\oint_{\gamma} G \cdot d\mathbf{r} = 0$ for every closed curve γ in U . therefore $f(p) = \int_{p_0}^p G \cdot d\mathbf{r}$ is path independent and $G = \nabla f \in W$. thus

$$[F] = \sum_{j=1}^8 a_j [F_j] \quad \text{and} \quad \boxed{\dim(V/W) = 8}.$$